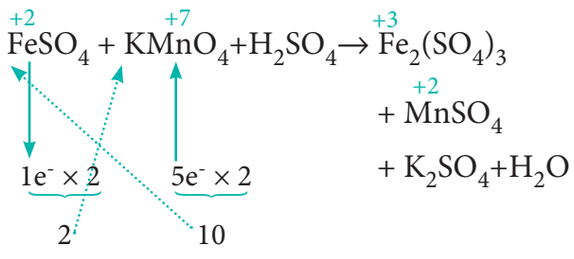


a) The oxidation number of Mn in $KMnO_4$ changes from +7 to +2 by gaining five electrons.

b) The oxidation number of Fe in $FeSO_4$ changes from +2 to +3 by loosing one electron.

Step 2

Since, the total number of electrons lost is equal to the total number of electrons gained, equate, the number of electrons, by cross multiplication of the respective formula with suitable integers on reactant side as below. Here, the product $Fe_2(SO_4)_3$ contains 2 moles of iron, So, the Coefficients $1e^-$ & $5e^-$ are multiplied by the number '2'



$10\ FeSO_4 + 2\ KMnO_4 + H_2SO_4 \rightarrow Fe_2(SO_4)_3 +$

$MnSO_4 + K_2SO_4 + H_2O$

Step 3 Balance the reactant / Product - Oxidised / reduced

Now, based on the reactant side, balance the products (ie oxidised and reduced).The above equation becomes

$10FeSO_4 + 2KMnO_4 + H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 2MnSO_4 + K_2SO_4 + H_2O$

Step 4 Balance the other elements except H and O atoms. In this case, we have to balance K and S atoms but K is balanced automatically.

Reactant Side : 10 'S' atoms (10 $FeSO_4$)

Product Side : 18 'S' atoms

$5Fe_2(SO_4)_3 + 2MnSO_4 + K_2SO_4$

$15S + 2S + 1S = 18S$

Therefore the difference 8-S atoms in reactant side, has to be balanced by multiplying H_2SO_4 by '8' The equation now becomes,

$10FeSO_4 + 2KMnO_4 + 8H_2SO_4 \rightarrow 5Fe_2(SO_4)_3 + 2MnSO_4 + K_2SO_4 + H_2O$

Step 5

Balancing 'H' and 'O' atoms

Reactant side '16'-H atoms ($8H_2SO_4$ i.e. 8 x 2H = 16 'H')

Product side '2' - H atoms (H_2O i.e. 1 x 2H = 2 'H')

Therefore, multiply H_2O molecules in the product side by '8'

$10\ FeSO_4 + 2\ KMnO_4 + 8\ H_2SO_4 \rightarrow$

$5\ Fe_2(SO_4)_3 + 2\ MnSO_4 + K_2SO_4 + 8H_2O$

The oxygen atom is automatically balanced. This is the balanced equation.

11th Std Chemistry 001-037.indd 25 23-12-2021 15:03:31